

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States 1.83 AWRT	AO1.1b	B1	1.83
(b)	Comments that David's statement is not or may be not true	AO2.3	R1	David's statement may not be true because
	Mentions that interval is based on mean from sample or Mentions that interval is for population mean length, not for population of lizard lengths	AO2.4	R1	Interval is centred on sample mean, not population mean or interval is for population mean length, not lizard lengths
(c)	Uses correct expression in b , s and n in equation relating to width value	AO1.1a	M1	$\frac{b \times s}{\sqrt{n}} = 4.27$ or $2 \times$ $\frac{b \times s}{\sqrt{n}} = 8.54$
	Solves for b FT incorrect s	AO1.1b	A1F	$\frac{b \times 4.15(45691)}{\sqrt{10}} = 4.27$ $b = 3.25$
	Deduces that $\beta = 99$ (condone 99 %)	AO2.2a	R1	From t tables, $\beta = 99$
				Total 6 marks

Q2.

- (a) Sample mean = 53.06, $s = 1.140$
Both. For s AWRT 1.14

B1

$$t_5 = 2.571$$

AWRT 2.57

B1

Sample mean $\pm t \times s / \sqrt{6}$
For $\sqrt{6}$

M1

Rest of formula. Allow $t_5 = 2.01$ to 2.02 , or $t_6 = 2.45$

m1

$$(53.06 \pm 1.20) = (51.86, 54.26)$$

Either form ± 0.01 in total.

A1

5

- (b) Sample mean is lower than last year's mean so claim **may** be true.
53.41 lies within c.i. so **not certain** that mean time is better.
Performance in competition does not depend on mean time.
Times seem to be improving.

E1 each for sensible comments either supporting or against statement up to a maximum of 2.

*Comment must be **uncertain**.*

E2

2

[7]

Q3.

(a) $\bar{x} = 948$ and $s^2 = 4817.25$

Both; AWRT 4820 ($s = 69.406$)

B1

$$t_8 = 2.896$$

AWRT 2.90

B1

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$$\text{C.I.} = 948 \pm 2.896 \times \sqrt{\frac{4817.25}{9}}$$

For division by $\sqrt{9}$

M1

For rest of expression, must be t_8 or t_9 ($= 2.821$)

m1

$$= 948 \pm 67.0 = (881, 1015)$$

Either form AWRT ± 67

Accept 1010 or 1020 as upper limit

A1

5

(b) (i) $(927 + 1063) \div 2 = 995$

CAO

B1

1

(ii) **Dependent on partial overlap**

Because of the overlap by the confidence intervals ...

E1

... no definite conclusion is possible

Accept "No evidence"

Edep1

2

SC Reference to evidence provided by the mean or the limits being lower 'suggesting' or 'providing evidence' or 'supporting' weight reduction scores 1

The statement must be not definite. Anything definite, eg. 'proves that' or 'shows that' scores 0

(E1)

[8]

Q4.

(a) $\bar{x} = \frac{\sum x}{n} = \frac{546}{15} = \frac{182}{5} = 36.4$

oe

B1

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{1407.6}{14} = 100.54 \text{ (or } s = 10.03)$$

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 $\sigma^2 = 93.84$ or $\sigma = 9.687$
 if $\frac{\sigma}{\sqrt{14}}$ used below

B1

$$t_{crit} = \pm 2.624$$

ignore signs for t_{crit} (allow $t = 2.62$)

*(if z used then **max** (B1B1B0 M0A0A0))*

B1

$$98\% \text{ CI for } \mu : 36.4 \pm 2.624 \times \frac{s}{\sqrt{15}}$$

$$\text{their } \bar{x} \pm t_{14} \times \frac{\text{their } s}{\sqrt{15}} \text{ or their } \bar{x} \pm t_{14} \times \frac{\text{their } \sigma}{\sqrt{14}}$$

M1

$$\left. \begin{array}{l} (29.6, 43.2) \\ 36.4 \pm 6.8 \end{array} \right\}$$

(allow any of the following for t_{14} :
1.345; 1.761; 2.145; 2.624; 2.977)

A1ft

$$= 29.6, 43.2$$

cao

A1

6

(b) $40.0 \in \text{C.I.}$

Must refer to 40 (dep M1)

E1ft

$\Rightarrow$ no change;

Dep on previous mark

E1ft

2

[8]

Q5.

(a)

$$\left. \begin{array}{l} Y \sim N(\mu_r, 4) \\ n = 16, \bar{y} = 450 \end{array} \right\}$$

(known variance) use $\Rightarrow z$

For 95% CI $z_{crit} = 1.96$

B1

$$\left. \begin{array}{l} 450 \pm 1.96 \times \frac{2}{\sqrt{16}} \\ 450 \pm 0.98 \end{array} \right\}$$

$$450 \pm 0.98$$

M1

$$(449, 451)$$

awrt

A1

3

(b) (i) $X \sim N(\mu_x, \sigma^2)$

(unknown variance) $\Rightarrow$ use t_{n-1}

$$\left. \begin{aligned} n=9 \quad \& \quad \bar{x} = \frac{4950}{9} = 550 \\ s^2_{n-1} = \frac{334}{8} = 41.75 \quad (s_{n-1} = 6.461) \end{aligned} \right\}$$

both

B1

For 90% CI $t_{crit} = 1.860$

B1

$$550 \pm 1.860 \times \frac{\sqrt{41.75}}{\sqrt{9}}$$

$\left\{ \begin{array}{l} \text{their } \bar{x} \pm t_8 \times \frac{\text{their } s_{n-1}}{\sqrt{9}} \\ \text{(must have at } t_8 \text{ - value)} \end{array} \right.$

M1

$$550 \pm 4.0$$

A1 ft

(546, 554)
awrt

A1

5

(ii) 545 not in 90% CI

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$\therefore$ Reject claim

Evidence to suggest that **mean** content of a bottle of tomato sauce is not 545 grams and hence to reject Holly's claim

Alternative

$$H_0 : \mu_x = 545$$

$$H_1 : \mu_x \neq 545$$

$$t = \frac{550 - 545}{\sqrt{41.75/9}} = 2.32$$

B1ft
(dep on (b)(i))

at 10% level of significance.

$$\left. \begin{aligned} t_{crit} = 1.86 < 2.32 \\ \text{Reject } H_0 \end{aligned} \right\} \quad \mathbf{B1}$$

Comment in context **E1**

E1 ft
(dep on (b)(i))

10% level of significance.

B1

3

Alternatives (such as):

Claim justified at 1% level of significance

$$t_{crit} = 3.355 > 2.32$$

Accept H_0

1% significance level

[11]

Q6.

(a) (i) $d^2 = \frac{93}{12}$

$$d = \sqrt{\frac{93}{12}} = \sqrt{7.75}$$

$$= 7.75$$

$$\Rightarrow d^2 = 7.75$$

M1

A1

2

(ii) 80% CI:

$$= 64.8 \pm 1.363 \times \sqrt{7.75}$$

$$t_{11} = 1.363 \text{ or } 1.36$$

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B1

$$= 64.8 \pm 3.79$$

$$64.8 \pm t_{11} \sqrt{7.75}$$

$$\text{iff } t_{11} = 1.363 \text{ or } 1.796$$

M1

$$= (61.0, 68.6)$$

AWRT

A1

3

(b) (i) $(64.8 - 5, 64.8 + 5)$

$$= (59.8, 69.8)$$

AWRT

B1

(ii) $w = 2\sqrt{7.75} \times t = 10$

M1

$$\Rightarrow t = 1.796$$

$$t = 1.79 \text{ to } 1.80$$

A1

$$P(X \geq 1.796) = 0.05$$

$$P(X \leq -1.796) = 0.05$$

iff $t = 1.796$ correct

M1

$$\Rightarrow P(|X| \leq 1.796) = 0.90$$

90% Confidence Level

A1

4

[10]

Q7.

$$\bar{x} = \frac{0.35}{10} = 0.035$$

$$s^2 = \frac{0.12705}{9} = 0.0141 \text{ or } (s = 0.1188)$$

both

B1

$$t_{\text{crit}} = 3.250$$

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B1

99% CI for μ :

$$0.035 \pm 3.25 \times \frac{\sqrt{0.0141}}{\sqrt{10}}$$

$$0.035 \pm 3.25 \times \frac{\sqrt{0.012705}}{\sqrt{9}}$$

M1

$$0.035 \pm 0.1221$$

*Iff $\bar{x}$, s and $t_9 = 3.25$
all correct in expression*

A1

$$(-0.087, 0.157)$$

CAO (3dp only)

A1

5

[5]

Q8.

(a) (i)
$$\bar{x} = \frac{1}{2}(70.65 + 80.35) = 75.5$$

AG

B1

1

(ii) Width of confidence interval
= $80.35 - 70.65$
= 9.7

B1

1

(iii) $t_{crit} = 2.602; v = 15$

B1

$$w = 2t \times \frac{s}{\sqrt{n}} \Rightarrow \frac{s}{\sqrt{n}} = \frac{9.7}{2 \times 2.602}$$

M1

Estimate of s.e = $\frac{s}{\sqrt{n}} = 1.86$
(1.864)

A1

3

(iv) Unbiased estimate of $\sigma^2 = 1.86^2 \times 16$

M1

= 55.6 (3sf)
AG (55.589)

A1

2

(b) 95% CI: $75.5 \pm 2.131 \times \frac{s}{\sqrt{n}}$

M1

= 75.5 ± 3.972
= (71.5, 79.5)



A1

2

(c) (i) (73.0, 78.0)

B1

1

(ii) $w = 2t \times \frac{s}{\sqrt{n}} \Rightarrow t = \frac{5}{2 \times 1.864} = 1.341$
 (AWFW 1.341 to 1.344)

M1

$$\Rightarrow \text{for } v = 15 \quad P(X \leq 1.341) = 0.90$$

$$\Rightarrow P(X \geq 1.341) = 0.10 \text{ and } P(X \leq -1.341) = 0.10$$

M1

$$\therefore P(|X| \leq 1.341) = 0.80$$

Percentage confidence interval = 80%

A1

3

[13]

Q9.

(a) (i) $\bar{x} = 43.5$

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$$s = 2 \quad (s^2 = 4)$$

B1

Assumption: Weights of boxes are normally distributed

B1

$$t_{0.975} = 2.365$$

B1

95% CI for μ :

$$\left. \begin{aligned} 43.5 \pm 2.365 \times \frac{2}{\sqrt{8}} \\ 43.5 \pm 1.6723 \end{aligned} \right\}$$

M1

$\Rightarrow (41.8, 45.2)$
 (AWRT)

A1

6

- (ii) CI contains mean (45)
*Must be clear use of 45 and **not** 43.5*

B1dep

Bishen's belief probably justified

B1dep

or

[Since 45 within CI] but close to upper
 limit, there is some evidence that Bishen's
 Belief is untrue
 [but the evidence is not significant at 5 %.]
 (75% of sample less than 45 grams)

(B1)

2

- (b) (i) $H_0 : \mu = 45$
 $H_1 : \mu < 45$
 (both)

B1

$$t = \frac{43.5 - 45}{\frac{2}{\sqrt{8}}}$$

Test statistic:

M1

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$$= -2.12$$

$$P(t_7 < -2.12) = 0.035791$$

$$< 0.05$$

A1

$$v = 7 \Rightarrow t_{\text{crit}} = -1.895$$

B1

$\Rightarrow$ Reject H_0

A1

Evidence at the 5% level of significance to support
 Abi's claim that **mean** content < 45 grams

E1

6

(ii) Type I error

B1

have/may have rejected H_0 when H_0 true
Clear statement

B1

or
 No error

(B1)

have/may have accepted H_0 when H_0 true
Clear statement

(B1)

2

[16]

Q10.

(a)

$$\bar{v} = \frac{1179}{10} = 117.9$$

B1

$$s^2 = \frac{1014.9}{9} = 112.8 \Rightarrow s = 10.6$$

$$\sigma^2 = 101.5 \quad (\sigma = 10.08)$$

B1

$$t_{0.995} = 3.250$$

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B1

99% confidence interval:

$$\text{or use of } \frac{\sqrt{101.5}}{3} = 3.359$$

$$117.9 \pm \frac{10.6}{\sqrt{10}} \times 3.250$$

$$(\text{their } \bar{v}) \pm \frac{(\text{their } s)}{\sqrt{10}} \times t_9$$

M1

$$= 117.9 \pm 10.9$$

$$(\text{on } \bar{x}, s \text{ and } t_9 = 3.25)$$

A1ft

$$\begin{aligned} &= (106.98, 128.82) \\ &= (107, 129) \\ &\quad \text{AWRT } (107, 129) \end{aligned}$$

A1

Assumption:
Speeds form a Normal Distribution

B1

7

- (b) John's claim is unlikely since 130 mph lies outside the confidence interval.

E1

1

[8]

Q11.

(a) (i) $\bar{x} = 3.19$ and $s^2 = \frac{1.849}{9} = 0.2054$
both ($s = 0.453$)

B1

$$t_9 = 3.250$$

B1

99% confidence interval:

$$\begin{aligned} &3.19 \pm 3.250 \times \frac{\sqrt{0.2054}}{\sqrt{10}} \\ &3.19 \pm (\text{their } t_9) \times \frac{\sqrt{0.2054}}{\sqrt{10}} \end{aligned}$$

M1

$$= 3.19 \pm 0.4658$$

A1ft

$$\begin{aligned} &= (2.72, 3.66) \\ &\quad (2.72 \text{ to } 2.73, 3.65 \text{ to } 3.66) \end{aligned}$$

A1

5

- (ii) Reasonable claim, with 3.5 within the

99% confidence interval
dep on correct CI in (a)(i)

B1E1
2

(b) $0.01 \times 200 = 2$

B1
1

[8]

Q12.

$\bar{x} = 65.375$

B1

$s = 3.662$ ($s^2 = 13.4$)

$\sigma = 3.426$ ($\sigma^2 = 11.7$)

B1

99% CI
 $t_7 = 3.499$

B1

$65.375 \pm \frac{3.662}{\sqrt{8}} \times 3.499$

$65.375 \pm \frac{3.426}{\sqrt{7}} \times 3.499$

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M1

$= 65.375 \pm 4.530$
 $= (60.8, 69.9)$

$(60.8 - 60.9, 69.90 - 69.91)$

A1F
5

[5]

Q13.

$\bar{x} = 39.5$ $s = 4.84$ ($s^2 = 23.4$)

$\sigma = 4.53$ ($\sigma^2 = 20.5$)

B1B1

$t_{\text{crit}} = 2.365$

B1

95% CI for μ

$$= \bar{x} \pm t_{\text{crit}} \times \frac{s}{\sqrt{n}}$$

$$= 39.5 \pm 2.365 \times \frac{4.84}{\sqrt{8}}$$

$$39.5 \pm 2.365 \times \frac{4.53}{\sqrt{7}}$$

M1

$$= 39.5 \pm 4.05$$

$$= (35.5, 43.5)$$

A1ft

5

[5]

Q14.

(a) $\bar{x} = 9.70$

$$\sigma = 0.0279$$

B1

$$s = 0.0294 \quad (s^2 = 8.67 \times 10^{-4})$$

$$(\sigma^2 = 7.8 \times 10^{-4})$$

B1

95% Confidence interval for μ :

$$= 9.70 \pm 2.262 \times \frac{0.0294}{\sqrt{10}}$$

$$\text{or } \frac{0.0279}{3}$$

B1M1

$$= 9.70 \pm 0.021$$

on their $\bar{x}$ and s

A1ft

$$= (9.68, 9.72)$$

A1

6

(b) (9.78, 9.82)

on their CI in (a)

B2ft

accept repeat of (a) using $(\bar{x} + 0.5)$ and s

2

[8]

Q15.

- (a) Assumption that the speeds of the cars passing through the village are normally distributed

B1

$$\bar{x} = 35.6$$

B1

$$s^2 = 38.27 \quad (s = 6.186)$$

$$(\sigma^2 = 34.44 \quad (\sigma = 5.869))$$

B1

99% Confidence Interval for μ

$$= 35.6 \pm 3.250 \times \frac{6.186}{\sqrt{10}}$$

or use of $\frac{\sqrt{34.44}}{3}$

B1

$$= 35.6 \pm 6.36$$

on their mean and standard deviation

M1A1ft

$$= (29.2, 42.0)$$

$$\text{CAO } (29.24, 41.96)$$

A1

7

- (b) Confidence interval includes 30 mph

B1ft

80% of sample exceed 30 mph limit

B1

Speed limit **not** adhered to

dependent on previous B1

B1

3

[10]

Q16.

(a) $\bar{x} = 301.0$

AWFW 24.4 to 24.5

B1

(b) $S^2 = (4.945)^2 = 24.5$

B1

2

$v = 9$

B1B1

$t = 3.250$

99% confidence interval for μ

$= 301.0 \pm 3.250 \times \left(\frac{4.945}{\sqrt{10}} \right)$

$(\text{their } \bar{x}) \pm t \times \left(\frac{\text{their } \sigma}{\sqrt{10}} \right)$

M1

$= 301.0 \pm 5.08$

A1

$= (295.9, 306.1)$

A1

5

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- (c) Since 290 is less than the lower limit of the confidence interval, it is probably safe to assume that the mean weight of packets of rice exceeds 290 grams

E2

2

[9]

Q17.

(a) $\bar{x} = 8.0$

B1

$S = 2.121$

B1

$V = 8$

B1

$$t = 1.860$$

(on their v)

B1ft

90% confidence interval for μ

$$= 8 \pm 1.860 \left(\frac{2.121}{3} \right)$$

M1

$$= 8 \pm 1.315$$

A1ft

$$= (6.68, 9.32)$$

(6.68 to 6.69, 9.31 to 9.32)

A1

7

- (b) The Headteacher's claim seems to be slightly optimistic
Headteacher's claim isn't supported by the evidence and

E1ft

because value of 5 outside the confidence interval

[It appears that the mean time to see a mathematics teacher is greater than 5 minutes]

E1ft

2

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[9]

Q18.

(a) $\bar{x} = \frac{254}{5} = 50.8$

B1

$$s = 4.55$$

B1

$$v = 5 - 1 = 4$$

B1

$$t_{\text{crit}} = 2.776$$

B1

95% confidence interval

$$= 50.8 \pm 2.7776 \times \frac{4.55}{\sqrt{5}}$$

ft their values

M1ft

$$= 50.8 \pm 5.648$$

$$= (45.2, 56.4)$$

A1

6

(b) 0.05

B1

1

[7]

Q19.

(a) $\Sigma x = 15.8$
 $\Sigma x^2 = 25.0592$

$$\bar{x} = \frac{15.8}{10} = 1.58$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{10}\right)$$

B1

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$$s^2 = \frac{25.0592}{9} - \frac{10}{9} (1.58)^2$$

$$= 0.01057$$

$$(s = 0.1028) \quad \text{AWRT } 0.011$$

B2

3

(b) 90% CI for μ

$$1.58 \pm \frac{s}{\sqrt{10}} \times 1.833$$

$$1.58 \pm 0.0596$$

M1A1ft

for v

B1

for t

B1ft

(1.52, 1.64)

for interval

A1ft

5

[8]

Q20.

(a) (i) $\bar{x} = 179.6$

CAO

B3

3

$s_x^2 = 133.16$

133 to 133.2

(B1 M1 A1 if by formula)

(ii) $v = 10 - 1 = 9$

B1

Critical value of $t = 1.833$

B1

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90% confidence limits for μ_x are

$$179.6 \pm 1.833 \sqrt{\frac{133.16}{10}}$$

(M1 only if z-value used)

M1

giving (172.9, 186.3)

ft on answers to (i) and t value

A1ft

AWFW (172.9 to 173, 186 to 186.3)

A1

5

(b) (i) Confidence limit for price in £ is

$$\left(1.20 \times \frac{173}{1000}, 1.20 \times \frac{186}{1000}\right)$$

M1

$$= (0.21, 0.22)$$

AWRT; or (21p, 22p); ft on CI in (a)(ii)

A1ft

2

- (ii) 20 pence is below confidence interval
so oranges are cheaper/ less profit per
orange

E1ft

comparing 20p with CI; ft on CI
sensible deduction
(both marks if lower end of CI implied)

E1

2

[12]

Q21.

(a) $\Sigma x = 324$ and $\Sigma x^2 = 13\,135.34$

$$\hat{\mu} = \bar{x} = \frac{324}{8} = 40.5$$

CAO

B1

$$\hat{\sigma}^2 = s^2 = \frac{1}{7} \left[13\,135.34 - \frac{324^2}{8} \right] =$$

use of; M0 for $\div 8$

M1

$$1.90 \text{ to } 1.91$$

AWFW

A1

3

- (b) (i) CI for μ is : $\bar{x} \pm \frac{t \times s}{\sqrt{n}}$
use of

M1

$$\text{Degrees of freedom, } v = 8 - 1 = 7$$

CAO

B1

95% $\Rightarrow p = 0.975$ so that $t = 2.365$
 AFWW 2.36 to 2.37

B1

$\therefore$ CI for μ is:

$$40.5 \pm \frac{2.365 \times \sqrt{1.90571}}{\sqrt{8}}$$

f.t. on (a)

A1f.t.

$\therefore 40.5 \pm 1.15$
 $\therefore (39.3 \text{ to } 39.4, 41.6 \text{ to } 41.7)$
 AFWW

A1

5

(ii) Distances are normally distributed.

B1

Distances are independent.
accept random sample

B1

EXAM PAPERS PRACTICE² [10]

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Q22.

(a) (i) $\hat{\mu} = \bar{x} = \frac{\Sigma x}{25} = \frac{88.2}{25}$
use of

M1

$= 3.528$
 CAO; accept 3.53

A1

$$\sigma^2 = s^2 = \frac{1}{24} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{25} \right) =$$

use of
[M0 for $\div 25$]

M1

$$\frac{1}{24} \left(326.2 - \frac{88.2^2}{25} \right) =$$

0.625 to 0.627

AWFW

A1

4

(ii) CI for μ is:

$$\bar{x} \pm \left(\frac{t}{z} \right) \frac{s}{\sqrt{n}}$$

use of

M1

$$t_{24}(0.995) = 2.797$$

CAO; accept 2.80
($z = 2.576$)

B1

$$\therefore 3.528 \pm 2.797 \sqrt{\frac{0.626}{25}}$$

f.t. on (i), (ii) & $\frac{t}{z}$

A1f.t.

$$\therefore 3.53 \pm 0.44$$

$$\therefore (3.09, 3.97)$$

AWRT

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A1

4

(b) As 99% CI **does not include** 4.1

f.t. on CI

B1f.t.

Mean weight of babies born in hospital is
different from/less than that in region

f.t.on CI; dep on B1f.t.

E1f.t.

2

[10]

Q23.

$$\bar{x} = 344.75 \quad s = 1.8323$$

$$344.75 \text{ (344.7 ~ 345)}$$

B1

1.8323 (1.83 ~ 1.835)

B1

90% confidence interval

$$344.75 \pm 1.895 \times \frac{1.8323}{\sqrt{8}}$$

7df

B1

1.895 (allow 1.89 or 1.9) their df

B1ft

Use of their $\frac{sd}{\sqrt{8}}$

M1

$$344.75 \pm 1.23$$

Completely correct method

m1

(343.52 , 345.98)

A1

343.52 (343.5 ~ 343.55) and
346.0 (345.9 ~ 346) allow 344 and 346
or 344.75 (344.7 ~ 344.8) ± 1.23 (1.225
~ 1.23)
Allow 345 ± 1

B1

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4,5 or 6sf given in final answer

[8]

Q24.

$$\bar{x} = 4.16667 \quad s = 0.5431 \quad \text{B1}$$

4.17 (4.16 to 4.17)

95% confidence interval for mean

0.543 (0.542 to 0.544) may be implied
by correct final answer

B1

$$4.16667 \pm 2.306 \times \frac{0.5431}{\sqrt{9}}$$

Use of their $\frac{s.d.}{\sqrt{9}}$

M1

8df

B1

2.306 - their df

B1

4.167 ± 0.147

completely correct method their t

m1

3.75 ~ 4.58

4.17(4.16 to 4.17) ± 0.417(0.417 to 0.418)
or 3.75(3.745 to 3.755) and 4.58(4.58 to 4.59)

□ A1

7

[7]

Q25.

(a) $\bar{x} = 135.6$

135.6 cao

B1

s = 0.45552

0.45552 (0.455 to 0.456) may be implied by correct interval

B1

95% confidence interval for mean is $135.6 \pm 2.306 \times \frac{0.45552}{\sqrt{9}}$
8df

B1

2.306

B1f.t.

Use of their $\frac{s}{\sqrt{9}}$

i.e. 135.6 ± 0.350

135.25 to 135.95

Correct method for interval

m1

Answer given to 4,5 or 6 sf, must be an interval

B1



135.25 (135.2 to 135.3) and
 135.95 (135.9 to 136)
Or 135.6 **cao** ± 0.35 (0.349 to 0.351)

A1

8

(b) $135.8 \pm 1.96 \times \frac{3.9}{\sqrt{60}}$
 1.96 **cao**

B1

Use of $\frac{3.9}{\sqrt{60}}$

B1

$$135.8 \pm 0.987$$

Correct method for interval

m1

$$134.81 \text{ to } 136.79$$

134.81 (134.8 to 134.820) and 136.79 (136.78 to 136.8)
Or 135.8 **cao** ± 0.987 (0.98 to 1)

A1

4

- (c) (i) Decrease
 Decrease

B1

1

- (ii) Increase - narrower interval $\rightarrow$ more likely to identify need
 for overhaul if $\mu \neq 135.0$
 Increase

E1

Reason

E1

2

- (d) Large variability as in (b) will lead to unsatisfactory production
 even if mean on target. Might prefer small variability as in
 (a) even if mean slightly off target.

Large variability unsatisfactory/makes deviation
 from 135 difficult to detect.

E1

Small variability with mean slightly off target
 may be preferable

E1

2

[17]

Q26.

(a) (i) $\bar{x} = 496.125$ $s = 7.318$
 $496.125(496 - 496.2)$

B1

$7.318 (7.31 - 7.33)$ - may be implied

B1

95% confidence interval for mean

Use of their $\frac{s}{\sqrt{8}}$ - generous

M1

$496.125 \pm 2.365 \times \frac{7.318}{\sqrt{8}}$
 Use of t
 Completely correct method } M2

m1

7df

B1

i.e. 496.125 ± 6.12
 $2.365 - \text{their df}$

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B1 ft.

$490.0 \sim 502.2$

$490.0 (489.95 - 490.05)$ and

$502.2 (502.2 - 502.3)$

Allow 502 or 496.125 ($496.1 - 496.2$)

$\pm 6.12 (6.1 - 6.12)$

A1

8

(ii) 80% confidence interval for mean
 Their t

M1

$496.125 \pm 1.415 \times \frac{7.318}{\sqrt{95}}$

i.e. 496.125 ± 3.66

492.5 ~ 499.8
 492.5 (492.4 – 492.5) and
 499.8 (499.7 – 499.9)
 Allow 492 and 500
 or 496.125 (496.1 – 496.2)
 ± 3.66 (3.65 – 3.67)

A1

2

(b) $499.6 \pm 1.57 \times \frac{9.3}{\sqrt{95}}$

Use of $\frac{9.3}{\sqrt{95}}$

Correct method, their z

m1

499.6 ± 1.57

$498.0 - 501.2$

498 (497.9 – 498.1) and
 501.2 (501.1 – 501.2)
 Allow 501 or 499.6 ± 1.57 (1.56 – 1.58)

A1

4

(c) (i) (A) 0.8
 0.8 cao

B1

(B) 0.15

Method

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M1

0.15 cao

A1

3

(ii) $0.95 \times 0.2 = 0.19$

Method

0.19 cao

A1

2

[19]

Q27.

(a) $\bar{x} = 48.4$ $s = 18.094$
 48.4 cao

18.094 (18.09 – 18.1)

B1 B1

99% confidence interval

Use of t

M1

$$48.4 \pm 3.250 \times \frac{18.094}{\sqrt{10}}$$

9 df
3.250 – their df

B1 B1ft

ie 48.4 ± 18.6

29.8 – 67.0

$\frac{s.d.}{\sqrt{10}}$
 Use of their
 Completely correct method – their t
 29.8(29.75 – 29.85)
 and 67.0 (66.95 – 67.05)
 or 48.4 and 18.6 (18.55 – 18.65)
 sc B1 B1 M1 M1 available if z used

M1 m1 A1

8

- (b) (i) 95% confidence interval
1.96 – allow t 1.979 – 1.984

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$$49.5 \pm 196 \times \frac{16.5}{\sqrt{110}}$$

$\frac{16.5}{\sqrt{110}}$
 Use of
 Their z

M1

49.5 ± 3.08

A1

4

46.42 – 52.58
 49.5 ± 3.08(3.08 – 3.09)
 or 46.42 (46.1 – 46.42)
 and 52.58(52.58 – 52.59)

Allow 46.4 and 52.6

- (ii) $49.5 \pm 1.96 \times 16.5$
Their z

M1

$17.2 - 81.8$
 49.5 ± 32.34 (32.3 – 32.4)
 or 17.2 (17.1 – 17.2) and
 81.8 (81.8 – 81.9)

A1

2

- (c) Mean greater than 25 but many
Mean $\neq$ 25

E1

individuals will be less than 25
Mean > 25
Some individuals < 25

E1 E1

3

- (d) Interval narrower
Interval narrower

E1

No need for normal assumption
Normal not needed
Or better estimate of σ , more representative, more
 reliable **not** more accurate

E1

2

- (e) Both samples random so valid suggestion
valid

E1

Would be better but not much

$\frac{16.5}{\sqrt{110}}$ very similar to $\frac{16.5}{\sqrt{120}}$
Better

E1

Not much - any three points

E1

3

[22]

Q28.

(a) $\bar{x} = 627$

cao

B1

$s = 640.38$

640 ~ 641

B1

95% confidence interval for mean is

9 df

B1

$627 \pm 2.262 \times 640.38 / \sqrt{10}$

for 2.262 or 2.26

attempt to use $s\sqrt{10}$

use of t-distribution

completely correct method–

allow incorrect t value

B1 M1 M1 M1

627 ± 458

$169 \sim 1085$

457 ~ 459

or 168 ~ 170 and 1080 ~ 1090. All rights reserved.

A1

8

(b) (i) 3 people drank no water

B1

suggesting a mode of zero which is a long way from the mean / not symmetrical

B1

(ii) amount cannot be negative

B1

3

(c) (i) $960 \pm 1.6449 \times 240 / \sqrt{135}$

for 1.6449 or 1.645 or 1.64 or 1.65 or 1.655~ 1.657



for use of $240/\sqrt{135}$
for completely correct method
(allow use of t-distribution)

B1 M1 M1

960 ± 34.0
 $33.5 \sim 34.5$

A1

4

$926 \sim 994$
 $925.5 \sim 926.5$ and $993.5 \sim 994.5$

(ii) first day/night shift unlikely to be typical of all employees

B1

1

[16]



EXAM PAPERS PRACTICE

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Examiner reports

Q2.

The calculation part was handled well, apart from by those students who used a z-value, and others who ignored the instruction to give the answer to two decimal places. A fair proportion of students recognised the uncertainty introduced because the 2011 mean lay within the 2012 confidence interval, and correctly deduced that the 2012 mean might lie below or above the previous year's mean. Others were far too definite in their conclusions. Strangely, a sizeable proportion of students equated a higher time with a better performance. Very few observed that success in races is not directly linked to mean time and depends also on other athletes' performances.

Q3.

The calculation part was well handled, apart from those students who used a z-value. A significant number performed a complicated calculation in part (b)(i), using the 2012 standard deviation and obtained the wrong answer. In part (b)(ii), a fair proportion of students recognised that the degree of overlap precluded any definite conclusion. Credit was given where students stated that there was some suggestion of a reduction in the mean, but others were often far too definite in their conclusions.

Q4.

Many students still seemed not to understand whether to use *t*-values or *z*-values in constructing confidence intervals or carrying out hypothesis tests. In this question, the sample size was *small* and was selected from a *normal population with unknown variance* so *t*-values should have been used. Unfortunately many students neither used a *t*-statistic nor did they give their answers to one decimal place as requested. Consequently many more students than expected failed to gain full marks. In part (b), marks were lost by those students who failed to state that 40 did or did not, depending on the confidence interval constructed, lie in the confidence interval. It was not sufficient to simply state "Mean in CI" since this could have referred to 36.4 rather than to the value of 40.

Q5.

In part (a), it was expected that candidates would understand that for a sample of any size from a normal population with known variance, the reference distribution is $N(0, 1)$. Far too many candidates based their choice of distribution on the small value of *n*, and consequently incorrectly used t_{n-1} as their reference distribution. This resulted in a loss of all marks for this part of the question.

In part (b)(i), most candidates correctly used a *t*-distribution. Some candidates still used the *z*-distribution, with $z_{\text{crit}} = 6449$ often seen.

Although most candidates correctly found $\bar{x} = 550$, there were far too many who thought that $s\sqrt{\frac{334}{9}}$ or, even worse, $s = \sqrt{334}$.

In part (b)(ii), far too many candidates failed to state the level of significance, even though this was specifically asked for in the question. Others incorrectly thought that the level of significance was 90%. Many candidates did not justify their comments by correctly stating that 545 was not in the confidence interval; incorrect statements, such as 'it is not in the CI', gained no credit.

Q6.

In part(a)(i), many candidates did not realise that an estimate for the standard error of

the mean was given by $d = \frac{s}{\sqrt{n}}$ and consequently that $d^2 = \frac{93}{12} = 7.75$. In part (a)(ii), where many correct answers were seen, some candidates used $z = 1.2816$ as their critical value whereas $t_{11} = 1.363$ was required. Others, confused by part (a)(i), often used $64.8 \pm$

$\frac{\sqrt{7.75}}{\sqrt{12}}$ rather than $64.8 \pm 1.363 \times \sqrt{7.75}$.

In part (b)(i), the correct answer was seen more often than not. Unfortunately, some candidates stated (60, 70) or (54.8, 74.8) as their answer, neither of which gained the mark available.

There were a lot of excellent solutions to part (b)(ii) from, in general, the more able candidates. The correct critical value of $t = 1.796$ often led to the correct answer of 90%. However, since candidates found that $t_{0.95} = 1.796$, some then quoted 95% as the answer. Unfortunately, some candidates insist on using z-tables for all calculations on confidence intervals (and indeed hypothesis testing in general) and consequently used $z = 1.79$ or $z = 1.80$ which, from tables, gave incorrect values of 92.8% or 92.6% respectively.

Q7.

Almost all candidates found that $\bar{x} = 0.035$ but their calculations of s or s^2 were less well done. Several candidates did not seem to understand how to find the required value from the given information, with $s = \sqrt{0.12705} = 0.3564$ often seen. Most of the candidates who found the correct value of s^2 usually correctly evaluated the required t -statistic.

Yet again, some candidates appeared not to understand that, when the value of the population standard deviation is unknown and the sample size is small, then t -values and **not** z -values should be used, otherwise significant marks will be lost. Too many candidates also ignored the request for the limits to be given to three decimal places.

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Q8.

In part (a)(i), although the great majority of candidates showed that the sample mean was

75.5 by simply evaluating $\frac{1}{2} (70.65 + 80.35)$, there were others who set up, then solved,

the pair of simultaneous equations $\bar{x} + 2.602 \times \frac{s}{4} = 80.35$ and $\bar{x} - 2.602 \times \frac{s}{4} = 70.65$, to

give $\bar{x} = 75.5$ and $s = 7.456$. This was usually a successful but lengthy method. In part (a)(ii), although most obtained the correct value of 9.7, some thought wrongly that the answer was 4.85. Part (a)(iii) was not answered well with many disappointing responses seen. Although some candidates calculated the correct value for the estimate of the

standard error of the mean by evaluating $\frac{s}{\sqrt{n}}$, the majority of those candidates who failed to obtain the correct answer thought that the standard error of the mean was equal to s and so gave 7.46 as their final answer. However, most candidates were able to obtain the correct answer to part (a)(iv), having already worked out the value of s in part (a)(i) or part (a)(iii).

There were also many good attempts at part (b), with the great majority of candidates evaluating $75.5 \pm 2.131 \times 1.864$ to give the correct interval. Unfortunately, some candidates insisted on using z-values, usually either 1.96 or 1.6449, and so lost credit.

Part (c)(i) was usually done very well with the correct interval often seen. There were also some excellent solutions to part (c)(ii), with $t = 1.341$ often seen followed by the correct answer of 80%. Unfortunately too many candidates wrongly thought that this value of t equated to the 90% level of confidence.

Q9.

Many excellent solutions were seen to part(a)(i), with $\bar{x} = 43.5$, $s = 2$ and $t = 2.365$ often used in the calculation of the 95% confidence interval as $43.5 \pm 2.365 \times \frac{2}{\sqrt{8}}$. Some candidates stated $s = 1.87$ or $s^2 = 3.5$, which only gained credit if used correctly in $43.5 \pm 2.365 \times \frac{1.87}{\sqrt{7}}$.

The vast majority of candidates either did not state an assumption or simply stated "It is normal" or "Sample is normal" or "Boxes are normally distributed". None of these gained any credit as it was the parent population or the weights of boxes of black peppercorns that were assumed to be normally distributed. In part (a)(ii), the comment was either too positive with candidates stating "Bishen's belief is true" or no reason was given as to why Bishen's belief may or may not have been justified.

In part (b)(i), the hypotheses were usually stated correctly, followed by a correct evaluation of the test statistic as -2.12 . However, far too many candidates stated that the critical value was $t = +1.895$ or $t = \pm 1.895$ and consequently went on to consider the wrong tail of the distribution, with the inevitable loss of credit. Part (b)(ii) was usually done very well, with only the occasional incorrect mention of a Type II error.

Q10.

In part (a), candidates were asked to state any assumption that they had made in constructing their confidence interval. Candidates should be aware that a meaningful interval cannot be constructed in this case unless they assume that they are dealing with a small sample from a normal population with unknown variance. It should also be noted that the assumption should be made with the context of the question in mind. Simply stating "Normal distribution" or "It is normal" or "the population is normal", without also stating what this population is, will not gain full credit. Also, many candidates incorrectly thought that the sample or the data had to be normally distributed.

Those candidates who did not state an assumption still calculated a confidence interval based upon the t -distribution. Most candidates calculated correctly that $\bar{v} = 117.9$ but were often less sure when it came to calculating an unbiased estimate, s^2 , for the variance of the population. Although a majority calculated correctly that $s^2 = \frac{1014.9}{9} = 112.8$ (or $s = 10.6$), there were many who thought wrongly that $s^2 = 101.49$ (or $s = 10.1$) or $s = \sqrt{1014.9} = 31.9$. Those who stated that $\sigma = 10.1$, nearly always went on to wrongly state that $\frac{10.1}{\sqrt{10}}$ was the value for the standard error. Appropriate use of the formulae booklet provided could have eliminated many of these errors.

The majority of candidates who calculated the correct values for $\bar{v}$ and s then used $t = 3.250$ and went on to construct the required confidence interval as (107, 129). Occasionally, the wrong value of t was extracted from the tables, and there were some candidates who thought that the reference distribution was $N(0, 1)$, rather than t_{n-1} ; this led them to use wrongly a variety of z -values in their calculations.

It seemed, from the responses to part (b), that very few candidates understand what a confidence interval shows. Many candidates firstly stated that John's claim was "wrong" and then indicated that this was because 99% of his serves were between 107 mph and 129 mph. Candidates should be aware that, when studying this section of the specification, emphasis should be given to both the construction and the interpretation of confidence intervals.

Q11.

This was, somewhat surprisingly, the worst answered question on the paper. A **small** sample from a normal distribution with **unknown** variance required use of the t -distribution. Consequently those candidates, of whom there were far too many, who used $z = 2.5758$ instead of $t = 3.250$ inevitably lost marks here. The better candidates

found $\bar{x} = 3.19$ and then correctly used $s^2 = \frac{1.849}{9} = 0.2054$ in constructing the 99%

confidence interval by evaluating $3.19 \pm t_9 \times \frac{\sqrt{0.2054}}{\sqrt{10}}$ to give (2.72, 3.66). Unfortunately,

some candidates, who thought incorrectly that $s^2 = \frac{1.849}{10} = 0.1849$ and so used $3.19 \pm t^9$

$\times \frac{\sqrt{0.1849}}{\sqrt{10}}$ instead of $3.19 \pm t^9 \times \frac{\sqrt{0.1849}}{\sqrt{9}}$, failed to find the correct confidence interval.

In part (a)(ii), a statement that '3.5 is within the confidence interval' with a conclusion that 'the claim was therefore reasonable' was all that was required here. Part (b) was usually done well, except by those candidates who thought that 1% was equivalent to finding

$\frac{1}{10} \times 200 = 20$.

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Q12.

This was a popular question, which resulted in the majority of candidates gaining full marks. Almost all of them realised that the data were a small sample from a normal distribution with unknown variance and therefore they had to use Student's t -distribution with 7 degrees of freedom to obtain the required critical value of $t = 3.499$.

Q13.

This question was very well answered by the majority of candidates, many of whom gained full marks. A few candidates persisted in using z -values when asked to construct a confidence interval for μ , whatever the given situation. However, this was not as prevalent as in previous papers.

Q14.

The confidence interval in part (a) was usually found correctly. However, on the whole, part (b) was not answered in the way that was expected. Most candidates attempted to

work out the confidence interval again using $\bar{x} = 9.6$ or 9.8 and $s = 0.0294$ by repeating the work already tested in part (a). Candidates were expected to write down their answer to part (b) from what they had already found in part (a) together with the information given in this part of the question.

Q15.

The numerical answers to part (a) were usually correct. However, there were many candidates who did not give an assumption or did not know what assumption they had made. The assumption that “The sample is normally distributed” was the usual false statement and “It is normally distributed” or simply “Normally distributed” were also insufficient. In fact it was the population from which the sample was taken that had to be assumed to have a normal distribution.

In part (b), a popular misconception by candidates seemed to be that the 99% confidence interval (29.2, 42.0) indicated that 99% of motorists travelled at speeds of between 29.2 mph and 42.0 mph, whereas it really indicated that there was a 99% confidence that the interval contained the true mean speed of all motorists travelling through the village. Whilst the majority of candidates indicated that 30 mph fell within the confidence interval, most then failed to realise that 80% of the sample had speeds in excess of 30 mph which consequently indicated that the speed limit was not being adhered to by most motorists.

Q16.

Most candidates were able to first show that $\bar{x} = 301.0$ and $s^2 = 24.5$. In part (b), although there were some good solutions seen, too many candidates used a z -value rather than the required t -value. Responses to part (c) were rather disappointing with comments often not in context and usually with no reference to the confidence interval that they had just found in part (b).

Q17.

There were many excellent solutions seen to part (a), with most candidates realising that they first had to find that $\bar{x} = 8.0$ and that $s = 2.121$. Although many candidates understood that the t -distribution had to be used with $\nu = 8$ and hence $t = 1.860$, since the data given formed a small sample from a normal distribution with unknown standard deviation, many others did not. These latter candidates usually used z -values from the normal distribution tables. Part (b) was not well answered as many candidates failed to make any reference to the context of the question or the confidence interval that they had just found.

Q18.

This question was answered very well indeed by the majority of candidates. Most realised that in part (a), as the data represented a small sample from a normal distribution with unknown variance, the use of a t -distribution with a critical value of $t = 2.776$ was required. However, some candidates thought that the appropriate critical value(s) should be $t = 2.132$ or even $z = \pm 1.96$ or $z = 1.6449$. Almost all candidates gained the mark available for part (b).

Q19.

Statistics 2A

In part (a), almost all candidates were able to calculate the value of $\bar{x}$, usually by inputting the given data into their calculators, and then using this as an unbiased estimate of the mean of the population. Unfortunately, many candidates lost credit by quoting $s = 0.1028$ instead of $s^2 = 0.0106$, the value asked for in the question. In part (b), it was pleasing to see that the majority of candidates understood that a small sample from a population with unknown variance suggested a t -distribution. Unfortunately, t (90%) tended to be read from tables rather than t (95%). It was only weak candidates who used z -values instead of the required t -values.

Statistics 2B

In part (a), most candidates were able to calculate the value of $\bar{x}$, usually by inputting the given data into their calculators, and then using this as an unbiased estimate of the mean of the population. Unfortunately, many candidates lost credit by quoting $s = 0.1028$ instead of $s^2 = 0.0106$, the value asked for in the question. In part (b), it was pleasing to see that most candidates understood that a small sample from a population with unknown variance suggested a t -distribution. Unfortunately, $t(90\%)$ tended to be read from tables rather than $t(95\%)$. Only a few candidates used z -values instead of the required t -values.

Q20.

Part (a) proved a good source of marks with a pleasing proportion of completely correct solutions. A small minority of candidates used a z -statistic instead of the appropriate t -statistic. Most candidates were able to deduce the required confidence interval in part (b)(i) and to use it in support of a sensible comment in part (b)(ii).

Q21.

The parts of this question leading to the confidence interval were well done. Calculations were accurate and presentation of the argument was generally logical and clear.

In part (b)(ii) a greater level of precision was required. It was the distances that the javelin was thrown that were assumed to be normally distributed and to be independent. Although candidates had some idea of what was required, their statements were not completely correct.

Q22.

All candidates determined correctly $\bar{\mu}$ as $\bar{x}$. However some candidates found a biased estimate of σ^2 ($\div n$) whilst others took the long route of calculating the sample variance and

then multiplying it by $\left(\frac{n}{n-1}\right)$. Calculations of the confidence interval in part (a)(i) were very sound, with the use of z , rather than t , the only error of note. All candidates knew what was required in part (b) and scored the 2 marks available.

Q23.

Most candidates obtained a correct solution to this question. Those who gave several alternative answers in order to ensure that one of them would be to an appropriate number of significant figures did not receive the last mark.

Q24.

This was intended as a straightforward opening question, but a surprisingly large proportion of candidates failed to gain full marks. The most common error was to use incorrect t -values. A few used z .

Q25.

Parts (a) and (b) were well answered. Unlike previous years, there was a mark in part (a) for giving the answer to an appropriate degree of accuracy (4, 5 or 6 significant figures were accepted). Those candidates who hedged their bets by giving alternative answers to different levels of accuracy did not receive this mark unless all their answers were to 4, 5 or 6 significant figures.

Most candidates made reasonable attempts at parts (c) and (d), although answers to part (d) revealed that many candidates did not realise that confidence intervals apply to means and are not intervals within which a given percentage of the population lies.

Q26.

The calculations posed relatively few problems, although a small minority used z where t was appropriate. There were many good answers to part (c) - a common error was to answer (c)(i)(B) as if the samples were independent, as they were in (c)(ii).

Q27.

Parts (a) and (b)(i) were well answered. The great majority of candidates successfully obtained standard deviation directly from their calculators as recommended. In part (a), a few candidates failed to use t , or used the wrong divisor for the standard deviation.

Only a minority of candidates were able to answer part (b)(ii) correctly, many merely repeated their answer to part (b)(i). Others calculated the width of the confidence interval.

The interpretation in part (c) was generally poor. Many candidates failed to make any connection between the confidence interval and the mean, believing rather that 95% of scores lie in the 95% confidence interval. The answers to parts (d) and (e) were better. In part (e), there was some concern as to the validity of combining two samples. However, as both samples were stated to be random samples the combined sample would also be a random sample.

Q28.

Most candidates were well prepared for part (a) although a few failed to use the t -distribution. Part (b) was not generally well answered, particularly part (b)(ii) where only a small minority pointed out that it is not possible to consume negative amounts of water and so the normal distribution cannot provide an exact mode.

All parts of (c) were well answered. It was expected that, as the sample was large, candidates would use z -values and the given standard deviation. Some used t -values.

Some interpreted the standard deviation as $\sqrt{\frac{\sum (x - \bar{x})^2}{n}}$ and applied a correction to obtain their value.

Most candidates made good attempts at parts (c)(i) and (c)(ii) although quite a few stated that because the sample was large the distribution (rather than the distribution of the sample mean) would be approximately normal.